

Discrete Linear Time-Invariant (LTI) Systems

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1 Lesson Objectives

2 Properties

3 Homework

Objectives

Properties

Homework



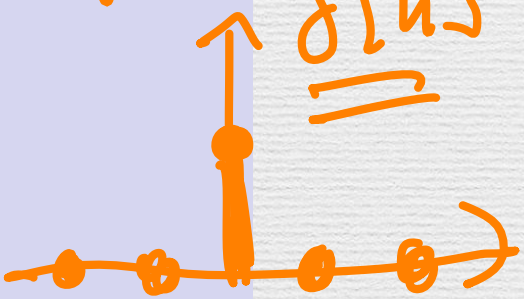
$$x[n] = \sum_{k=-\infty}^{+\infty} x[k]\,\delta[n-k]$$



$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]\,h[n-k]$$

Convolution Sum :

$$y[n] = x[n] * h[n]$$



Objectives

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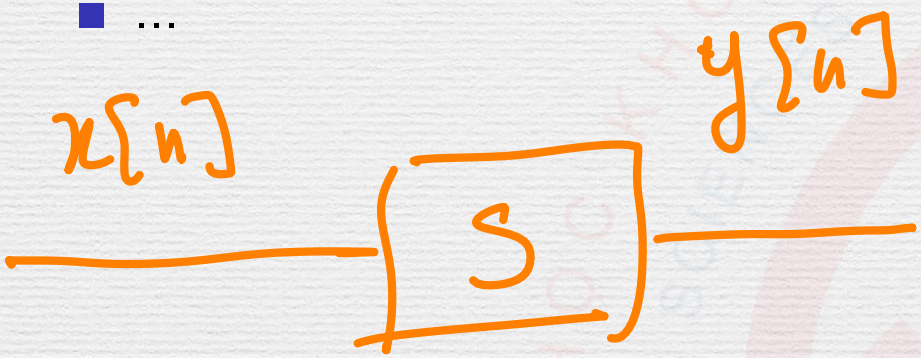
Lesson Objectives

At the end of this lesson, you should be able to

Lesson Objectives

At the end of this lesson, you should be able to

1. determine (and justify) properties of a LTI system:
   - Causality
   - Stability
   - ...



- $y[n]$ at time $n_0 \Rightarrow$ we need only $x[n]$ at time $n_1 \leq n_0 \Rightarrow$ Causality
- if $\|x[n]\| \leq N \Rightarrow \|y[n]\| \leq M$, $\exists M$ that

1 Lesson Objectives

2 Properties

- Commutative
- Distributive
- Associative
- Invertibility
- Causality
- Stability

Computation Properties

Systems Properties

3 Homework

Commutative

Definition

$$x[n] * h[n] = h[n] * x[n]$$

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k] \cdot h[n-k]$$

$$= h[n] * x[n] = \sum_{k=-\infty}^{+\infty} h[k] \cdot x[n-k]$$

proof this link is true

Commutative

Definition

$$x[n] * h[n] = h[n] * x[n]$$

Verify this property.

$$x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k] \cdot h[n-k]$$

$$\begin{cases} t = n-k \\ k = n-t \end{cases}$$

$$x[n] * h[n] = \sum_{t=n-(+\infty)}^{t=n-(-\infty)} x[n-t]\, h[t]$$

$$= \sum_{t=-\infty}^{+\infty} h[t]\, x[n-t] = h[n] * x[n]$$

Distributive

Definition

$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$$

$$= \sum_{k=-\infty}^{+\infty} x[k]\left(h_1[n-k] + h_2[n-k]\right)$$

$$= \sum_{k=-\infty}^{+\infty} \left(x[k]\,h_1[n-k] + x[k]\,h_2[n-k]\right)$$

$$= \sum_{k=-\infty}^{+\infty} x[k]\,h_1[n-k] + \sum_{k=-\infty}^{+\infty} x[k]\,h_2[n-k]$$

$$= x[n] * h_1[n-k] + x[n] * h_2[n-k]$$

Distributive

Objectives

Properties

Commutative

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Causality

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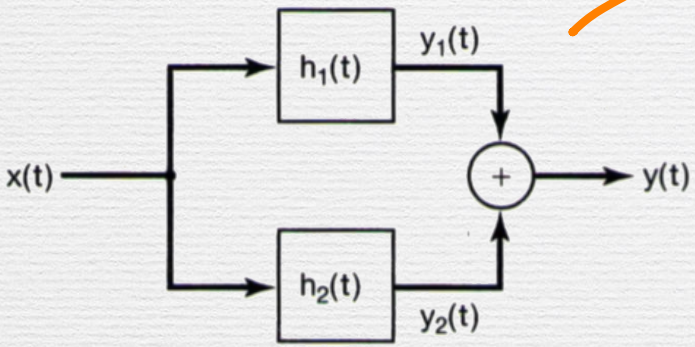
Homework

Definition

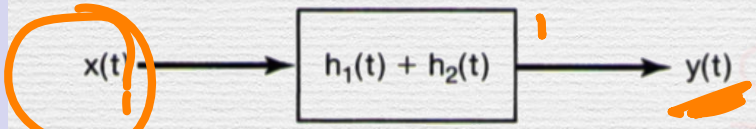
$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$$

Verify this property.

Distributive - Parallel Interconnection



(a)



(b)

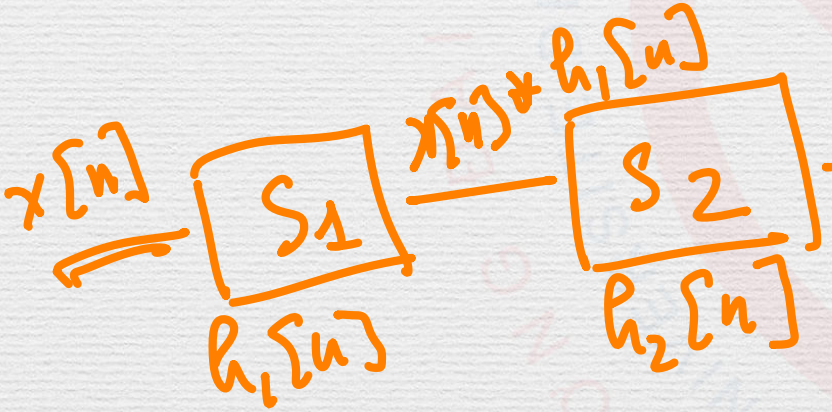
$$y[n] = x[n] * h_1[n] + x[n] * h_2[n]$$

$$= x[n] * [h_1[n] + h_2[n]]$$

$$h[n]$$

expensive

$$y[n] = (x[n] * h_1[n]) * h_2[n]$$



Associative

Definition

$$x[n] * (h_1[n] * h_2[n]) = (x[n] * h_1[n]) * h_2[n]$$

$h[n]$

verify:

$$(x[n] * h_1[n])[n] = \sum_{k_1=-\infty}^{+\infty} x[k_1]\, h_1[n-k_1]$$

$A[n]$

Associative

Definition

$$x[n] * (h_1[n] * h_2[n]) = (x[n] * h_1[n]) * h_2[n]$$

Verify this property.

$$x[n] * h_2[n] = \sum_{k_2=-\infty}^{+\infty} x[k_2]\, h_2[n-k_2]$$

$$= \sum_{k_2=-\infty}^{+\infty} \left(\sum_{k_1=-\infty}^{+\infty} x[k_1]\, h_1[k_2-k_1] \right) h_2[n-k_2]$$

$$= \sum_{k_2=-\infty}^{+\infty} \sum_{k_1=-\infty}^{+\infty} x[k_1]\, h_1[k_2-k_1]\, h_2[n-k_2]$$

$$\sum_{k_1=-\infty}^{+\infty} x[k_1] \left[\sum_{k_2=-\infty}^{+\infty} h_1[k_2 - k_1]\, h_2[n - k_2] \right]$$

$$\underbrace{h_1[n] * h_2[n]}_{\beta[n]} = \sum_{k_2=-\infty}^{+\infty} h_1[k_2]\, h_2[n - k_2]$$

$$\beta[n - k_1] = \sum_{k_2=-\infty}^{+\infty} h_1[k_2]\, h_2[n - k_1 - k_2]$$

$$= \sum_{k_2=-\infty}^{+\infty} h_2[k_2]\, h_1[n - k_1 - k_2]$$

Boundary changing

Associative - Series Interconnection

Signals & Systems

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Objectives

Properties

Commutative

Distributive

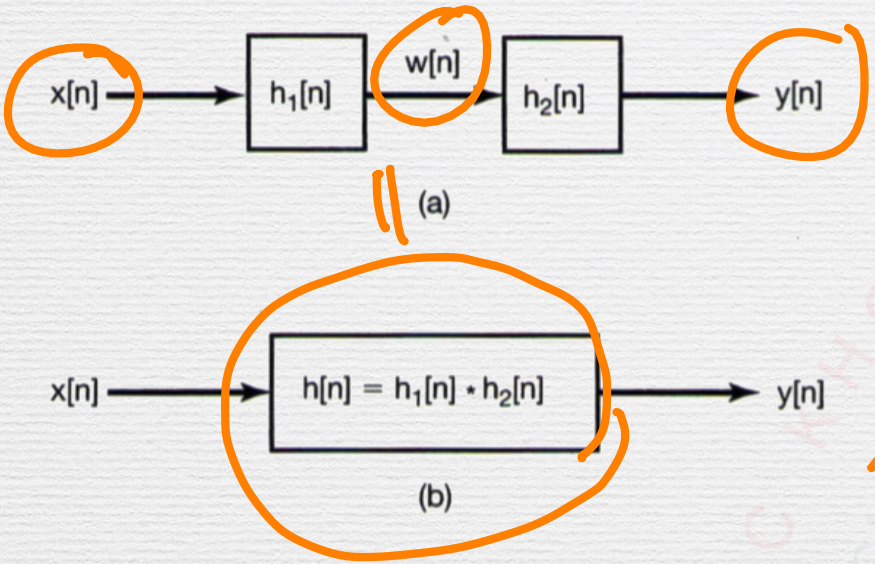
Associative

Invertibility

Causality

Stability

Homework

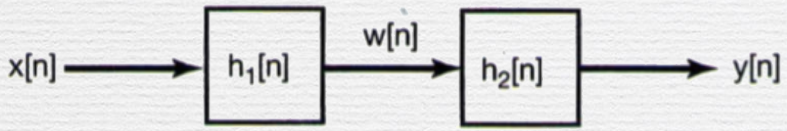


$$(x[n] * h_1[n]) * h_2[n]$$

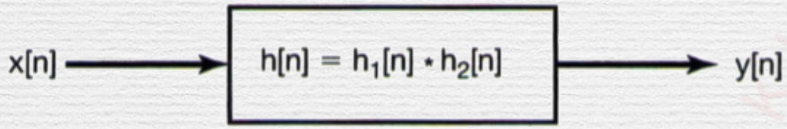
$$= x[n] * (h_1[n] * h_2[n])$$

$$h[n]$$

Associative & Commutative



(a)



(b)

Associative & Commutative

Signals & Systems

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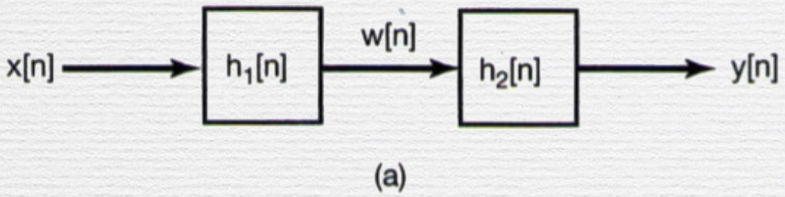
Associative

Invertibility

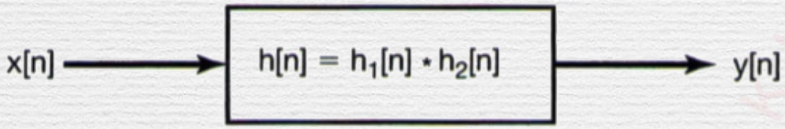
Causality

Stability

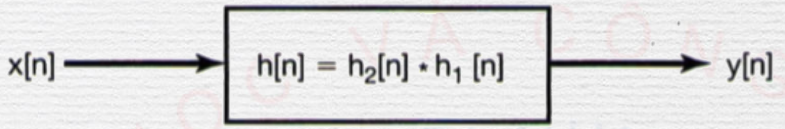
Homework



(a)



(b)



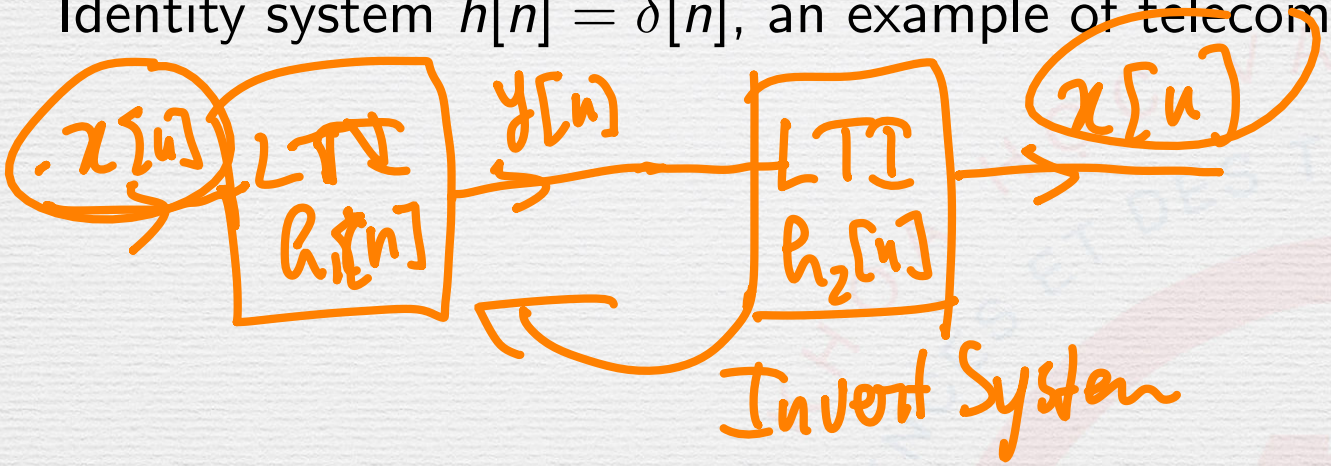
(c)



(d)

Invertibility

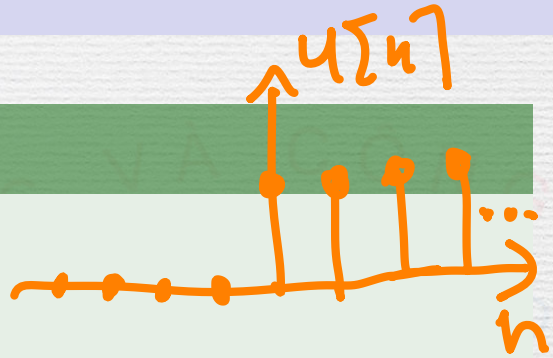
Identity system $h[n] = \delta[n]$, an example of telecommunication.



$$y[n] = x[n] = (x[n] * h_1[n]) * h_2[n]$$

$$x[n] = x[n] * \frac{(h_1[n] * h_2[n])}{\delta[n]}$$

Invertibility



Example

Given a system with impulse response

$$h_1[n] = u[n]$$

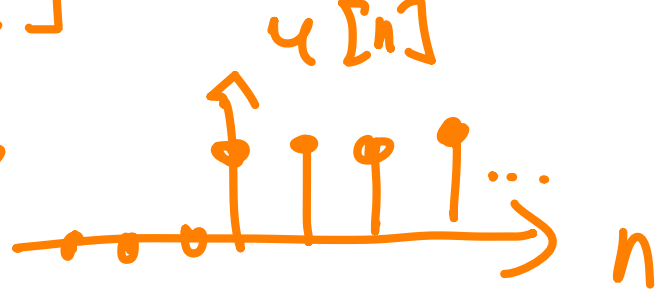
Verify that its inverse system has $h_2[n] = \delta[n] - \delta[n-1]$

$h_1[n] * h_2[n] = \delta[n]$

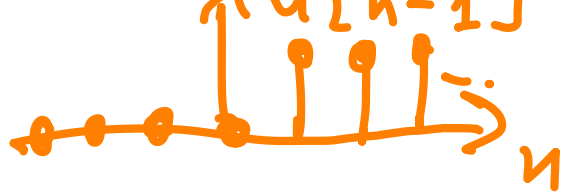
$$h_1[n] * h_2[n] = \sum_{-\infty}^{+\infty} u[k]\,(\delta[n-k] - \delta[n-k-1])$$

$$= \underbrace{\sum_{k=0}^{+\infty} \delta[n-k]}_{A[n]} - \underbrace{\sum_{k=0}^{+\infty} \delta[n-k-1]}_{B[n]}$$

$$A[n] = \begin{cases} 1 & \text{if } n \geq 0 \\ 0 & \text{if } n < 0 \end{cases} \Rightarrow$$



$$B[n] = \begin{cases} 1 & \text{if } n \geq 1 \\ 0 & \text{if } n < 1 \end{cases} \Rightarrow$$



$$h_1[n] * h_2[n] = A[n] - B[n] = u[n] - u[n-1]$$

$$h_1[n] * h_2[n] = \delta[n]$$

Causality

Signals & Systems

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Objectives

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Associative

Invertibility

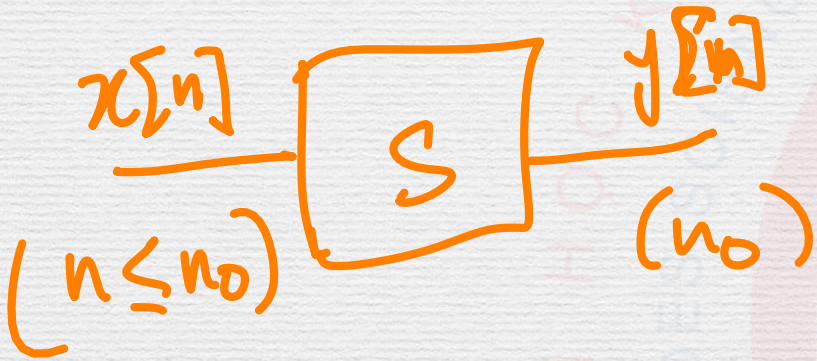
Causality

Stability

Homework

A LTI system is causal iff its impulse response satisfy the condition:

$$h[n] = 0 \text{ for } n < 0$$



$$y[n_0] = \sum_{k=-\infty}^{+\infty} x[k]\, h[n_0 - k]$$

$$(k \le n_0)$$

Stability

Signals & Systems

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Homework

A LTI system is stable iff its impulse response is absolutely summable

$$\sum |h[n]| < \infty$$

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]\,h[n-k]$$



if $\exists N$ that $\|x[n]\| \leq N$

then $\exists M$ that $\|y[n]\| \leq M$

$\Rightarrow S$ is stable

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Homework

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]\,h[n-k]$$

Discrete LTI Systems Exercises

2.3, 2.4, 2.7, 2.24

4 steps to perform convolution Sum
(Apply only for Discrete signals that can represented in vector)

- S1: Fixed 1 signal ($x[n]$)
- S2: Reverse IR. $\Rightarrow \overleftarrow{h}[n] = h[-n]$
- S3: Align Reverse Signal with Fixed one
  - ↳ Element wise multiple $\Rightarrow$ Sum
- S4: Assign value to $y[0] \Rightarrow y[n_0]$

shift Reverse Signal (n_0 steps)

$x[n] = [1, \underset{\uparrow}{2}, 2, 3]$

$h[n] = [1, \underset{\uparrow}{2}, 2] \Rightarrow \hat{h}[n] = [2, \underset{\uparrow}{2}, 1]$

$y[0] = [1, \underset{\uparrow}{2}, 2, 3] = 8$

$[2, \underset{\uparrow}{2}, 1]$

$y[-1] = [1, \underset{\uparrow}{2}, 2, 3] = 4$

$[2, 2, \underset{\uparrow}{1}]$

$\vdots$

Conclude: $y[n] = [1, 4, \underset{\uparrow}{8}, 11, 10, 6]$

- Disc LTI system that can represent in vector
- Don't Miss value and original arrow

2.4)

$x[n] = [0, 0, 0, 1, 1, 1, 1, 1, 1]$

$h[n] = [0, 0, 0, 0, \underbrace{1, \ldots, 1}_{12}]$

$y[0] = [0, 0, 0, 0, (1), \ldots, 1] = 0$

$[1, 1, 1, 1, (0), 0, 0, 0]$

$y[1] = \ldots = 0, \; y[2] = 0 \ldots$

$y[7] = 1$

$y[8] = 2$

$y[9] = 3$

$\vdots$

$y[12] = 6$

$y[13] = 6$

$\vdots$

$y[18] = 6$

$y[19] = 5$

$\vdots$

$y[23] = 1$

$y[24] = 0$

$\Rightarrow$ Conclusion: don't try this way

2.3)

$$y[n] = \sum_{k=-\infty}^{+\infty} \left(\frac{1}{2}\right)^{k-2} u[k-2]\, u[n-k+2]$$

$$= \sum_{k=2}^{n+2} \left(\frac{1}{2}\right)^{k-2}$$

$$= 4 \sum_{k=2}^{n+2} \left(\frac{1}{2}\right)^{k}$$

$$= \dots$$